Legal Protection of Digital Property

Lecture 9

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Ownership

- patent belongs to the inventor: s.9E

- if $\geq$ 2 inventors, then collectively own the patent and each is entitled to an equal share as "tenants in common" in the patent: s.54(1)

- each co-owner may
  - license, assign or mortgage his share only with the consent of all others: s.54(3)
  - otherwise exploit the patent for his own benefit without the consent of others: s.54(2)

NB. Co-owners' rights can be varied by contract between themselves.

Employee invention: s.57

- employee invention belongs to the employer in any of the following situations:
  - within the employee's normal or specifically assigned duties AND an invention might reasonably be expected from carrying out the duties

– within the employee's duties and the employee has a special obligation to further the interests of the employer's undertaking because of the nature of his duties and his particular responsibilities

NB. Slightly different from the position under copyright law.

E1: X, a technician in ChipTech, a chip manufacturing company, came up with an idea for a new chip while trouble-shooting a faulty chip. He has applied for a patent for the new chip. If the patent is granted, who will be the owner?

E2: Similar to E1. But X is the chief engineer of ChipTech.

Patentable Invention: s.9A

- 4 conditions:
  - susceptible of industrial application
  - new ("novel" or "not anticipated")
  - involves an inventive step ("not obvious")
  - not excluded under ss.9A(2)-(6)

NB. Novelty and inventiveness assessed against "state of the art".

Susceptible of Industrial Application: s.9D

- can be used in any kind of industry, including agriculture ie. invention must have a useful or practical purpose

NB. Certain matters deemed not susceptible of industrial application eg. medical treatment of human or animal body: s.9A(4).

Novelty: s.9B

- "new" if the invention does not form part of the "state of the art"

- "state of the art" comprises
  - everything made available to the public (in HK or elsewhere) whether by means of a written or oral description, by use or any other way **before the "material date"** of the patent application

NB. "Material date" = filing date of the patent application or, if priority is claimed, the priority date: PO, s.2(1).

NB. Must ensure that any disclosure of invention before applying for patent does not enable a skilled person in the field to make the invention.

NB. Non-prejudicial disclosure with regard to novelty: s.11A.

Inventiveness: s.9C

- "inventive" if having regard to the state of the art, the invention is "not obvious to a person skilled in the art" (ie. someone who knows the state of the art)

NB. Assessment of novelty and inventiveness requires expert evidence.

- obvious inventions in the past
  - mere collocation of known devices
    - » *Williams v Nye* (1890) 7 RPC 62

- analogous use
  - » *Thermos v Isola* (1910) 27 RPC 388

- mere verification of previous predictions
  - » *Genentech v Wellcome Foundation* [1989] RPC 147

NB. If an invention is known in the state of the art, a slightly different invention may be novel, but may fail for being obvious (ie. not inventive).

Excluded Subject-Matter

- s.9A(2): not regarded as "inventions":
  - discovery, scientific theory or mathematical method
  - aesthetic creation
  - scheme, rule or method for performing a mental act, playing a game or doing business, or a **computer program**
  - presentation of information

- BUT exclusion applies "only to the extent to which a patent or patent application relates to such subject-matter or activities as such"

NB. Similar provisions in UK and European Patent Convention.

- true meaning of "as such" not easy to ascertain

- UK's and EPO's interpretation of "as such" has changed over the years!

NB. See exclusions under ss.9A(4)-(6) for subject matters other than computer programs.

UK's Early Interpretation of "As Such"

- a computer program by itself (ie "as such") is not patentable.
- What about a computer program stored in a ROM? cf. *Gale's Application* [1991] RPC 305 (CA).

Q: What about a program loaded into a known computer ie. known computer + new program?

NB. UK's early position on software patent was very different from the US. US Patents Act does not exclude computer programs from patentability

- **US Patents Act, s.101: "Whosoever invents or discovers any new and useful process, machine, manufacture, or composition of matter, or any new and useful improvement thereof, may obtain a patent therefor, …"**

Subsequent development in UK (influenced by EPO)

- computer program can be patented if the program makes a "technical contribution" to the state of the art ie.
  - solves a technical problem;
  - uses technical means; AND
  - produces technical effect which is observable/measurable eg. reduced energy consumption, improved product quality

Cf. *Merrill Lynch* [1989] RPC 561 (CA)

"Technical Effect"

- the normal interaction between a program and the hardware when the program is run cannot constitute the requisite technical effect
- there must be further effects deriving from running the program which have a technical character or which solve a technical problem
- Cf. *IBM's Application* [1999] RPC 861 (EPO TBA) **(*RR)**

- effect of a program can be
  - external eg. program-controlled manufacturing process, OR
  - internal eg. program-controlled memory management thereby improving the performance of the computer

NB. Computer program can be claimed on its own! Opened the door for patenting software despite the exclusion.

- BUT the **net effect** of the invention must not itself be excluded by the law eg.
  - if **net effect** is business method, excluded: *Merrill Lynch* [1989] RPC 561 (CA) (a computerized trading system in stocks)
  - if **net effect** is performing mental act, excluded: *Wang's Application* [1991] RPC 463 (a system enabling an expert to store his knowledge in a hierarchical form so that expert advice could be offered)

E1: A new operating system that features a novel method of data buffering which increases the speed of file transfer on the Internet by 30%. Excluded from patentability?

E2: A ROM containing a computer program which implements a new method of solving differential equations.

NB. The above principles as regards patenting software have extended to all excluded subject matters in s.9A(2).

Cf. Latest 4-step test in *Aerotel v Telco Holdings* [2006] EWCA Civ 1371:

1. properly construe the claim;
2. identify the actual contribution;
3. ask whether it falls solely within the excluded subject matter;
4. check whether the actual or alleged contribution is actually technical in nature

NB. The *Aerotel* test is essentially a reformulation of the earlier tests on exclusion stemming from *Merrill Lynch* and *IBM.* The test is now applied to all subject matters in s.9A(2)

PRC's Position

- computer programs by themselves are excluded as "rules and methods for mental activities" which are not patentable: PRC Patent Law, Art 25(2)

- BUT if a program is combined with hardware to form an "integrated technical solution" that solves a technical problem and produces technical effect, the combination is patentable

Recommended Reading

- IBM's Application [1999] RPC 861 (EPO TBA)